

# A 2D Gaussian–Mixture Task: Eight Gaussians on a Circle (and its Ellipse Misspecification)

tabpfm-misspecification

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## 1 Model

We use the 2D instance (`gaussian_mixture_2d`) of the `GaussianMixtureHD` task, so the parameter  $\theta$  and the data  $x$  both live in  $\mathbb{R}^2$  ( $D = 2$ ). The well-specified generative model is a *Gaussian prior with a Gaussian-mixture likelihood*, which is conjugate and therefore yields a closed-form posterior.

**Prior.**

$$\theta \sim \mathcal{N}(0, \sigma_p^2 I_D), \quad \sigma_p = 5. \quad (1)$$

This is a single broad isotropic Gaussian — there is no mixture in the prior.

**Anchors (the “circle”).** We fix  $K = 8$  anchor points evenly spaced on a circle of radius  $\rho = 8$ :

$$c_k = (\rho \cos \varphi_k, \rho \sin \varphi_k), \quad \varphi_k = \frac{2\pi k}{K}, \quad k = 0, \dots, K-1. \quad (2)$$

**Likelihood / data-generating process.** Draw a discrete component  $z$ , shift  $\theta$  by the corresponding anchor, and add isotropic Gaussian noise:

$$z \sim \text{Uniform}\{0, \dots, K-1\}, \quad x \mid \theta, z = \theta + c_z + \varepsilon, \quad \varepsilon \sim \mathcal{N}(0, \sigma^2 I_D), \quad \sigma = 0.5. \quad (3)$$

Marginalizing over the component  $z$  gives the mixture likelihood

$$p(x \mid \theta) = \frac{1}{K} \sum_{k=0}^{K-1} \mathcal{N}(x; \theta + c_k, \sigma^2 I_D). \quad (4)$$

So a data point is the parameter  $\theta$  teleported to one of the eight ring positions, plus a little noise.

## 2 Closed-form posterior

Each term of (4) is Gaussian in  $\theta$  and the prior is Gaussian, so by conjugacy (applied per component) the posterior is again a  $K$ -component Gaussian mixture:

$$p(\theta \mid x) = \sum_{k=0}^{K-1} w_k \mathcal{N}(\theta; \mu_k, \tau^2 I_D), \quad (5)$$

with shared (shrunk) variance, per-component means, and weights

$$\tau^2 = \frac{\sigma_p^2 \sigma^2}{\sigma_p^2 + \sigma^2}, \quad \mu_k = \frac{\tau^2}{\sigma^2} (x - c_k), \quad w_k \propto \exp\left(-\frac{\|x - c_k\|^2}{2(\sigma_p^2 + \sigma^2)}\right). \quad (6)$$

The reference observation is set to the symmetric point  $x_{\text{obs}} = 0$ . Then every  $\|c_k\| = \rho$  is equal, so all weights are equal,  $w_k = 1/K$ , and the means reduce to

$$\mu_k = -\frac{\tau^2}{\sigma^2} c_k, \quad (7)$$

i.e. *eight equal-weight, isotropic blobs* (standard deviation  $\tau$ ) sitting on a circle of radius  $\frac{\tau^2}{\sigma^2} \rho$ . With the chosen constants  $\tau^2 = \frac{25 \cdot 0.25}{25.25} \approx 0.248$  ( $\tau \approx 0.498$ ) and  $\frac{\tau^2}{\sigma^2} \approx 0.990$ , so the modes lie at radius  $\approx 7.9$  with blob width  $\approx 0.5$  — well separated. Empirically the eight posterior modes land at radius 7.92 with relative spread 0.2%, evenly spaced every  $45^\circ$ .

Because the posterior is an explicit Gaussian mixture, reference samples are drawn *exactly* (no MCMC): pick  $k \sim \text{Categorical}(w)$ , then  $\theta = \mu_k + \tau \xi$  with  $\xi \sim \mathcal{N}(0, I_D)$ .

### 3 Misspecification: circle $\rightarrow$ ellipse

The misspecified simulator (`ellipse_modes`) keeps the *same* process but moves the anchors onto an ellipse with semi-axes  $(\rho, \rho r)$ , where  $r = \text{axis\_ratio}$ :

$$c_k^{\text{miss}} = (\rho \cos \varphi_k, \rho r \sin \varphi_k), \quad x = \theta + c_z^{\text{miss}} + \sigma_x \varepsilon. \quad (8)$$

Setting  $r = 1$  recovers the circle; the default  $r = 0.5$  squashes the ring vertically. Observations are then generated from the elliptical layout while inference still assumes the circular reference posterior (6) — that geometric mismatch is the misspecification under study.

## 4 Visualizations

### 5 Posteriors for other observations $x$

The point  $x = 0$  is special only because it is equidistant from all anchors. For a general observation the posterior is still the eight-component mixture (6), but two things change. From  $\mu_k = \frac{\tau^2}{\sigma^2} (x - c_k)$ , the eight modes still lie on a circle of radius  $\approx \rho$ , now *re-centered* at  $\approx x$ . From  $w_k \propto \exp(-\|x - c_k\|^2 / 2(\sigma_p^2 + \sigma^2))$ , the weights *tilt* toward the modes whose anchor  $c_k$  is nearest to  $x$ . Here the denominator  $\sigma_p^2 + \sigma^2 \approx 25.25$  is moderate, so the tilt is pronounced: e.g. at  $x = 0$  all  $w_k = 0.125$ , while at  $x = (6, 6)$  the weights range over  $[0.002, 0.48]$  — a couple of modes dominate and the far ones all but vanish. Figure 4 illustrates this for four observations.

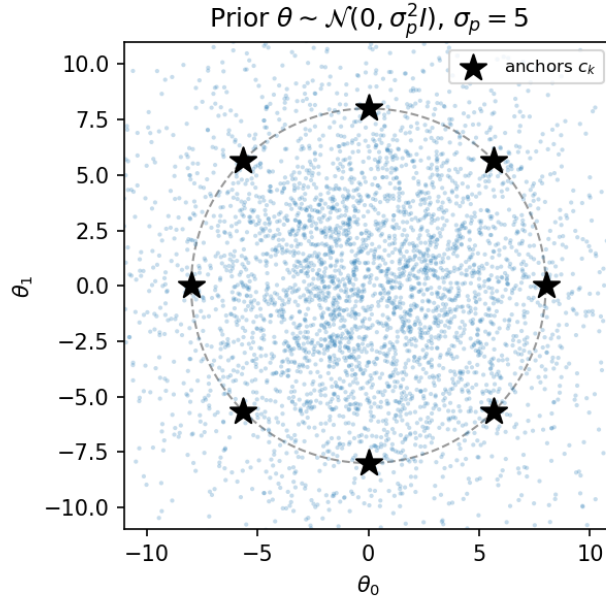


Figure 1: Prior  $\theta \sim \mathcal{N}(0, \sigma_p^2 I)$  (blue), a single broad isotropic Gaussian. Black stars mark the eight anchors  $c_k$  on the dashed circle of radius  $\rho = 8$ .

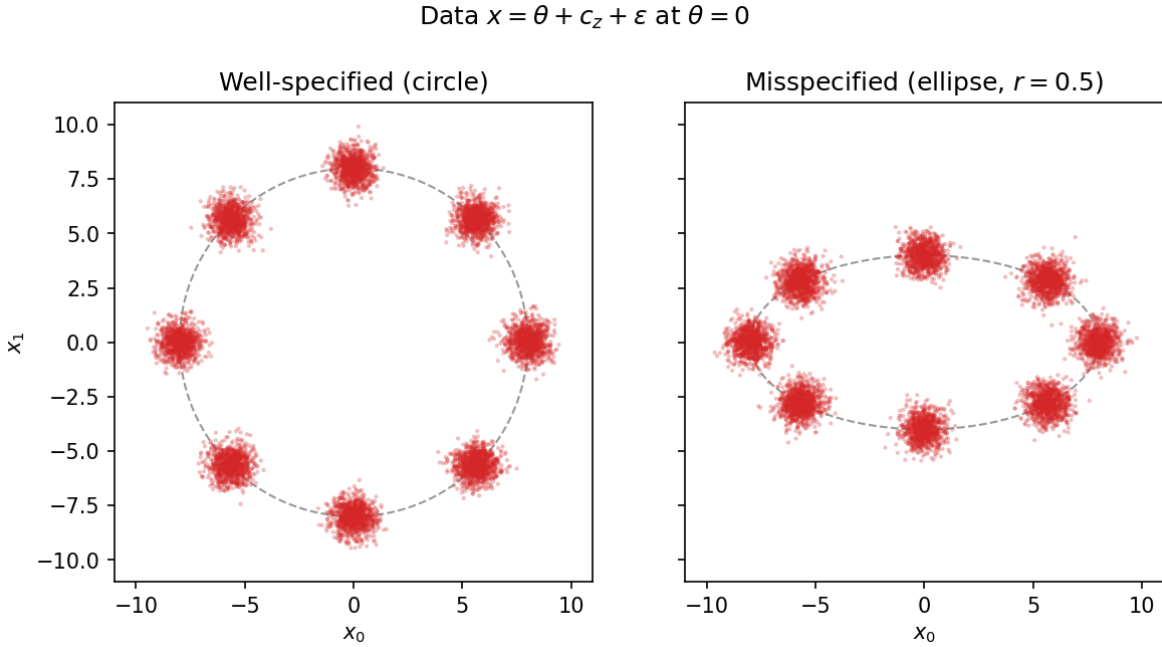


Figure 2: Conditional data distribution  $x \mid \theta = 0$ . *Left*: well-specified model — the eight noisy clusters sit on the dashed circle. *Right*: misspecified model with `axis_ratio= 0.5` — the same clusters now sit on an ellipse.

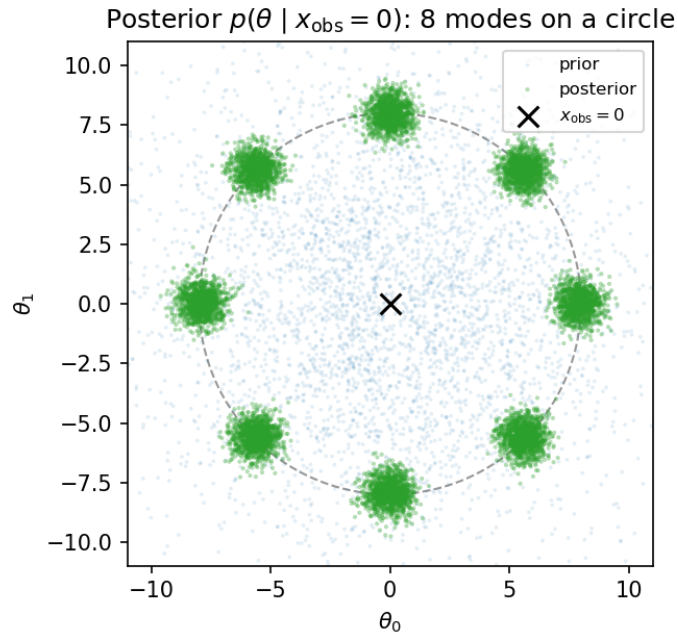


Figure 3: Closed-form posterior  $p(\theta \mid x_{\text{obs}}=0)$  (green): eight equal-weight Gaussian modes on a circle, sharply concentrated relative to the prior (faint blue). The observation  $x_{\text{obs}} = 0$  is the symmetric point (black cross).

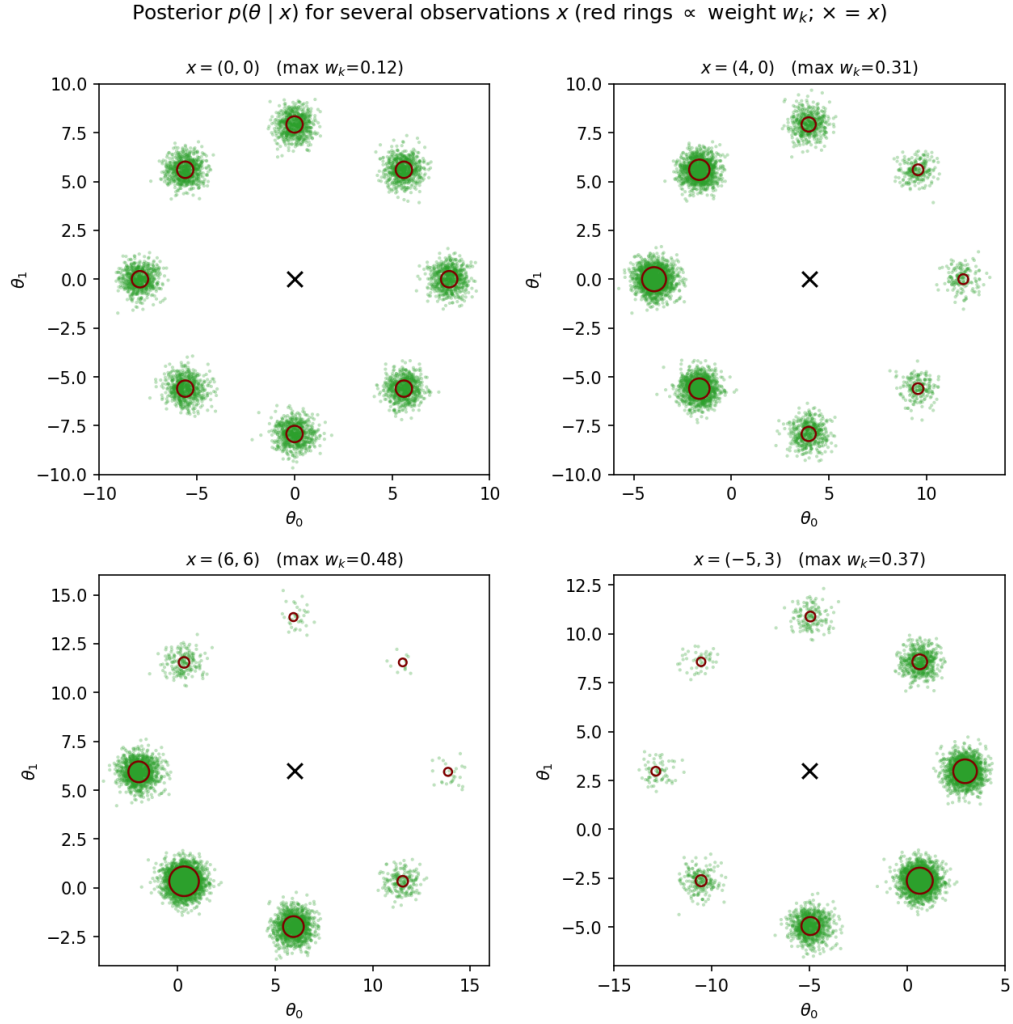


Figure 4: Posterior  $p(\theta | x)$  (green samples) for four observations  $x$  (black  $\times$ ). Red rings mark the closed-form mode means  $\mu_k$ , with area proportional to the mixture weight  $w_k$ . As  $x$  moves, the ring of eight modes re-centers near  $x$  and the weights tilt toward the modes nearest  $x$  (the per-panel maximum  $w_k$  is shown in each title).